

Mengxiao Gao

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Cornell University

Ph.D. Student, ORIE

Research Interests

Statistical foundations of generative models: diffusion and score estimation, differential privacy, and learning from model-generated data.

Education

Cornell University

Ph.D. in Operations Research and Information Engineering

Advisor: Yuchen Wu

Aug. 2025 – Present

GPA: 4.0+/4.0

Tsinghua University

B.S. in Mathematics and Physics

Double major in Civil Engineering and Systems

Sept. 2021 – June 2025

GPA: 3.8/4.0

Current Research

Private Score Estimation and Diffusion Trajectories

Investigating the statistical limits of differentially private score estimation for diffusion models, with a focus on minimax risk. Exploring whether diffusion semigroup structure and dependence across noise levels can enable efficient estimation of the full score trajectory under privacy constraints.

Model Collapse and Synthetic-Data Feedback

Studying recursive generative learning when model-generated samples feed into subsequent training, focusing on how finite-sample and approximation errors accumulate across generations.

Publications & Selected Research

MixCIT: A Kernel Based Local-Polynomial Debiased Test for Conditional Independence on Mixed-Type Data

Mengxiao Gao, Kyra Gan, Promit Ghosal. *Preprint, 2026*. [arXiv]

- Developed a kernel-based conditional independence test for mixed data using local-polynomial debiasing.
- Established asymptotic null distributions across data-type combinations and a dimension-free $n^{-1/4}$ detection threshold under local alternatives, with near-quadratic algorithms and analytic calibration.

Hybrid Top-Down Global Causal Discovery with Local Search for Linear and Nonlinear Additive Noise Models

Sujai Hiremath, Jacqueline R.M.A. Maasch, Mengxiao Gao, Promit Ghosal, Kyra Gan.

Advances in Neural Information Processing Systems (NeurIPS), 2024. [Paper]

- Developed a hybrid causal discovery method combining topological ordering with local nonparametric search for linear and nonlinear additive-noise models.
- Established correctness and worst-case polynomial-time guarantees; evaluated the method on synthetic data.

Technical Skills & Awards

Programming: Python, PyTorch, R, C/C++, SQL.

Awards: Cornell Fellowship 25'.